

# Aravind Kannappan

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[LinkedIn](#) | [GitHub](#) | [Website](#)

Open To Relocation  
US Citizen

## EDUCATION

### New York University

MS in Applied Statistics  
*Specialization in Machine Learning*  
Aug 2024 – Aug 2026

### Baylor University

BS in Statistics  
*Minor in Biology*  
Aug 2020 – Aug 2024

## TECHNICAL SKILLS

### Languages:

Python, TypeScript, SQL, R, C++

### AI/ML Frameworks:

LangChain, LangGraph, PyTorch, Hugging Face, vLLM, Scikit-learn, RAG, Prompt Engineering, Tensorflow, MCP

**Frontend:** React, REST APIs

### Cloud & DevOps:

AWS (S3, EC2, Bedrock), GCP, Docker, Kubernetes, CI/CD, Git

## COURSEWORK

- NLP, Deep Learning, Data Engineering, Distributed Systems, Big Data, Bayesian Methods, Biostatistics
- Data Structures & Algorithms, Linear Algebra, Calculus, Database Systems, System Design, Operating Systems

## TECHNICAL WRITING [BLOG](#)

- Wrote posts on optimization, generalization, and probabilistic modeling from first principles.
- Built reproducible Python notebooks, training scripts, and benchmarks.
- Studied failure modes (leakage, shift, calibration) and proposed fixes.

## PUBLICATIONS

- How ViT Represents Position (ICLR 2027)**  
Mechanistically compared APE vs RoPE to study how ViTs encode and represent spatial position.
- Layerwise Test of PRH in Astronomy (NeurIPS, 2026)**  
Tested PRH across 36 models; shared geometry did not imply shared physical representations. [Link](#)

## CERTIFICATIONS

Google Cloud Professional MLE, AWS MLE & Solutions Architect Associates, NVIDIA Associate: Gen AI & LLMs

## SUMMARY

Two time founder with production experience designing LLM powered systems, RAG pipelines, and multi agent orchestration across diverse domains. Published researcher and open source contributor advancing work in AI safety and evaluation. Strong foundation in Machine Learning Systems and software engineering, working across the full stack in fast moving, ambiguous environments

## WORK EXPERIENCE

### Research Engineer Fellow [Github](#)

Eleuther AI | Summer of Open AI Research

Jun 2026 – Present

Remote

- Applied SAEs and linear probes on residual-stream activations across 24 transformer layers to localize galaxy-morphology emergence, reaching 0.91 probe accuracy using Pytorch
- Computed mutual k-NN alignment (k=10) across 4 foundation models on 120K astronomical samples to test Platonic Representation Hypothesis via cosine similarity and rank correlation
- Built activation patching and feature ablation for causal interventions over 16K-feature SAE dictionaries; SLURM multi-GPU pipeline (8 GPUs) cut large-scale feature extraction 6x
- Trained crosscoders across 4 models to isolate shared galaxy-decomposition features; validated inter-model agreement at 0.89 F1 on held-out probing classifiers
- Built layer-wise FAISS-indexed feature-extraction pipeline over 2M galaxy embeddings; KL-divergence shift detection pinpointed representation-emergence thresholds across 24 layers

### Cofounder, Software Engineer AI/ML [Website](#) | [GitHub](#)

Replays AI | AI-Powered Immersive Post-Game Recaps

Sep 2025 – May 2026

New York, NY

- Designed ingestion pipeline writing 5M+ structured play-by-play events from the ESPN API into PostgreSQL as system of record; stored raw video referenced by pointer, cutting compute by 40%.
- Enforced schema validation & player disambiguation across 50K+ records in Python, cutting 95% duplicate entries; anchored video segmentation & CV inference for accurate highlight alignment
- Orchestrated 3 parallel agentic inference pipelines via asyncio spanning event feature extraction, Claude Vision play classification (14 labels, confidence-scored), and reducing inference by 35%.
- Decoupled storage and inference from the frontend via 12+ FastAPI REST endpoints; implemented Redis recap caching, sustaining sub-second responses on cache hits at 85% hit rate
- Built React 18 and TypeScript frontend with Vite and TanStack Query rendering personalized game feeds and fan-perspective AI recaps, reducing load time by 28% via component memoization.

### Founder and ML Engineer [Website](#) | [GitHub](#)

Synthure | Mult-Agent Clinical Intelligence

Jul 2023 – Jul 2025

New York, NY

- Built hybrid retrieval over 1.43M ICD-10 embeddings (pgvector cosine + BM25 fallback, IVFFlat-indexed) for clinical code grounding, reaching MRR@5 of 0.91 on offline benchmarks
- Architected multi-agents communicating through typed IR data classes, confidence-routing high-risk cases (score  $\geq 50$ ) to Claude Sonnet and the rest to Haiku with an async dispatcher
- Trained a Gradient Boosting denial predictor on 38,924 clinical transcriptions (AUC-ROC 0.87) to triage claims and gate LLM escalation upstream of generation
- Enforced 3-tier autonomy guardrails in code with forced tool\_use schemas and post-generation citation validation, eliminating fabricated facts (2.3% drift) across 60K+ records
- Cascaded NER stack at 94.2% on held-out MTSamples, served through 42 endpoints across 4 portals; deployed on Supabase with JWT auth and HIPAA PHI audit logging

## OPEN SOURCE

### PodBench [Website](#) | [GitHub](#)

LLM-agent benchmark on deterministic, resettable SQL environments; KEDA-autoscaled K8s (2–24 replicas), per-run token metering,  $\sim 10\times$  cheaper via prompt caching, cost-vs-reward frontier models

### Popper [Website](#) | [GitHub](#)

Breaks faulty proof specs via counterexamples pre-verification, cutting proof-search compute  $\sim 20\%$ ; 176/178 unfaithful specs caught (99% recall, 0 false alarms, F1 0.99)

### Context Forge [Website](#) | [Github](#)

Pareto benchmark for LLM context compression (GPT-4o/4/2, SQuAD/GovReport):  $\sim 92\%$  quality at  $\sim 51\%$  fewer tokens vs. 71% truncation; droppability classifier at 0.96 ROC-AUC

### VeriGrad RL [Website](#) | [Github](#)

Mechanistic-interpretability framework for safety RL: policy learns verifier-scored activation interventions; With OOD jailbreak/over-refusal evals (0% jailbreak, 0% over-refusal)

## PROJECTS

### MOSAIC [Website](#) | [Github](#)

Bayesian disease-surveillance system across 39 countries; validated on 1,334 forecasts (AUROC 0.917)

### Covera [Website](#) | [Github](#)

Insurance engine; MEPS Monte-Carlo simulation over 1,600 CMS plans, rank by risk-adjusted cost

### RobustSight: Advancing AI Safety and Alignment [Github](#)

CV AI-safety framework; saliency training raised PGD robustness & IoU to 45% at under 3% loss